CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

# 2023 ANNUAL REPORT

PREPARED FOR:

ALVES & SON DAIRY #2

13831 EAST AVE

TURLOCK CA 95380

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Alves and Son Dairy #2

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 13831 East Ave. AVE | Turlock | Merced | 95380 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/01/1985

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X090-X004-XXXX X041-X110-X027-XXXX X041-X110-X028-XXXX

**B. OPERATORS**

Alves, Joe

Operator name: Alves, Joe

Telephone no.: Landline: ; Cellular: (209) 585-8221

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 13745 Newport RD | Turlock | CA | 95380 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Bettencourt, Joe Louie

Legal owner name: Bettencourt, Joe Louie

Telephone no.: Landline: (209) 668-3276; Cellular: (209) 620-3400

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 14011 East AVE | Turlock | CA | 95380 |

Bettencourt, Mary Jean

Legal owner name: Bettencourt, Mary Jean

Telephone no.: Landline: (209) 668-3276; Cellular: (209) 602-0042

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 14011 East AVE | Turlock | CA | 95380 |

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 40 | 0 | 0 |
| Number under roof | 990 | 210 | 110 | 0 | 0 | 0 |
| Maximum number | 990 | 210 | 110 | 40 | 0 | 0 |
| Average number | 990 | 210 | 110 | 40 | 0 | 0 |
| Avg live weight (lbs) | 1,400 | 1,500 | 1,100 | 775 | | |

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 30,898.24 *tons per reporting period*

Total nitrogen from manure: 395,220.11 *lbs per reporting period*

After ammonia losses (30% loss applied): 276,654.08 *lbs per reporting period*

Total phosphorus from manure: 65,633.22 *lbs per reporting period*

Total potassium from manure: 194,147.14 *lbs per reporting period*

Total salt from manure: 514,431.00 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 31,440,000 *gallons*

Total nitrogen generated: 111,146.39 *lbs*

Total phosphorus generated: 23,485.08 *lbs*

Total potassium generated: 223,081.88 *lbs*

Total salt generated: 889,693.83 *lbs*

| | | |
|---|---|---|
| | 31,440,000 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 31,440,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Domestic Well | Ground water |
| Irrigation Well | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

*No commercial or other nutrient imports entered.*

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 1 | 164 | 164 | 2 | both | X041-X110-X002-XXXX<br>X041-X110-X027-XXXX |
| Field 2 | 135 | 135 | 2 | process wastewater | X041-X100-X019-XXXX<br>X041-X110-X028-XXXX |
| Field 3 | 25 | 25 | 2 | both | X041-X100-X015-XXXX<br>X041-X100-X019-XXXX |
| Totals for areas that were used for application | 324 | 324 | 6 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 324 | 324 | 6 | | |

## B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 164 Plant date: 10/13/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/03/2023 | 3,817.20 *ton* | Dry-weight | | 69.8 | 12,320.00 | 2,820.00 | 25,640.00 | | 8.50 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 23.28 | 173.20 | 39.64 | 360.46 | 1,194.97 |

## Field 1

### 05/25/2023: Corn, silage

Crop: Corn, silage  Acres planted: 164  Plant date: 05/25/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/18/2023 | 4,978.00 *ton* | Dry-weight | | 68.3 | 13,600.00 | 2,225.00 | 13,550.00 | | 5.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.35 | 261.72 | 42.82 | 260.76 | 1,019.94 |

## Field 2

Field name: Field 2

### 10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 135  Plant date: 10/13/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/01/2023 | 2,656.30 *ton* | Dry-weight | | 69.6 | 13,450.00 | 3,050.00 | 26,100.00 | | 9.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 19.68 | 160.90 | 36.49 | 312.24 | 1,148.47 |

### 05/25/2023: Corn, silage

Crop: Corn, silage  Acres planted: 135  Plant date: 05/25/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/29/2023 | 4,097.80 *ton* | Dry-weight | | 69.3 | 12,150.00 | 1,950.00 | 14,950.00 | | 7.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.35 | 226.44 | 36.34 | 278.63 | 1,323.26 |

## Field 3

Field name: Field 3

### 10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 25  Plant date: 10/13/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/01/2023 | 491.90 *ton* | Dry-weight | | 62.6 | 12,500.00 | 3,100.00 | 23,900.00 | | 10.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 19.68 | 183.97 | 45.62 | 351.75 | 1,471.76 |

### 05/25/2023: Corn, silage

Crop: Corn, silage  Acres planted: 25  Plant date: 05/25/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/29/2023 | 758.80 *ton* | Dry-weight | | 71.4 | 12,500.00 | 1,900.00 | 18,700.00 | | 6.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.35 | 217.02 | 32.99 | 324.66 | 1,145.85 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Field 1 - 10/13/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/13/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/25/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 100.00 | 16.84 | 263.78 | 978.20 | 4,320,000.00 *gal* |
| Irrigation Well | Ground water | 11.33 | 0.00 | 0.00 | 0.00 | 19,200,000.00 *gal* |
| Application event totals | | 111.33 | 16.84 | 263.78 | 978.20 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 11.22 | 0.00 | 0.00 | 0.00 | 19,008,000.00 *gal* |
| Application event totals | | 11.22 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 100.29 | 23.56 | 188.07 | 792.57 | 5,280,000.00 *gal* |
| Irrigation Well | Ground water | 11.05 | 0.00 | 0.00 | 0.00 | 18,720,000.00 *gal* |
| Application event totals | | 111.34 | 23.56 | 188.07 | 792.57 | |

### Field 1 - 05/25/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/25/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## Field 1 - 05/25/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/08/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 104.84 | 33.22 | 91.86 | 0.00 | 1,230.00 *ton* |
| Application event totals | | 104.84 | 33.22 | 91.86 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 05/31/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 34.19 | 8.03 | 64.11 | 270.19 | 1,800,000.00 *gal* |
| Irrigation Well | Ground water | 15.30 | 0.00 | 0.00 | 0.00 | 25,920,000.00 *gal* |
| Application event totals | | 49.49 | 8.03 | 64.11 | 270.19 | |

| | | | | |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 15.64 | 0.00 | 0.00 | 0.00 | 26,496,000.00 *gal* |
| Application event totals | | 15.64 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 34.19 | 8.03 | 64.11 | 270.19 | 1,800,000.00 *gal* |
| Irrigation Well | Ground water | 14.90 | 0.00 | 0.00 | 0.00 | 25,248,000.00 *gal* |
| Application event totals | | 49.09 | 8.03 | 64.11 | 270.19 | |

| | | | | |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 14.73 | 0.00 | 0.00 | 0.00 | 24,960,000.00 *gal* |
| Application event totals | | 14.73 | 0.00 | 0.00 | 0.00 | |

## Field 1 - 05/25/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 37.55 | 8.17 | 59.53 | 240.43 | 1,500,000.00 *gal* |
| Irrigation Well | Ground water | 15.30 | 0.00 | 0.00 | 0.00 | 25,920,000.00 *gal* |
| Application event totals | | 52.85 | 8.17 | 59.53 | 240.43 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 15.87 | 0.00 | 0.00 | 0.00 | 26,880,000.00 *gal* |
| Application event totals | | 15.87 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 37.55 | 8.17 | 59.53 | 240.43 | 1,500,000.00 *gal* |
| Irrigation Well | Ground water | 14.45 | 0.00 | 0.00 | 0.00 | 24,480,000.00 *gal* |
| Application event totals | | 52.00 | 8.17 | 59.53 | 240.43 | |

## Field 2 - 10/13/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/13/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/28/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 89.42 | 15.06 | 235.89 | 874.74 | 3,180,000.00 *gal* |
| Irrigation Well | Ground water | 13.80 | 0.00 | 0.00 | 0.00 | 15,720,000.00 *gal* |
| Application event totals | | 103.22 | 15.06 | 235.89 | 874.74 | |

## Field 2 - 10/13/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 13.69 | 0.00 | 0.00 | 0.00 | 15,600,000.00 *gal* |
| Application event totals | | 13.69 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.23 | 20.49 | 163.56 | 689.30 | 3,780,000.00 *gal* |
| Irrigation Well | Ground water | 13.59 | 0.00 | 0.00 | 0.00 | 15,480,000.00 *gal* |
| Application event totals | | 100.81 | 20.49 | 163.56 | 689.30 | |

## Field 2 - 05/25/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/25/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 38.77 | 9.11 | 72.69 | 306.35 | 1,680,000.00 *gal* |
| Irrigation Well | Ground water | 20.01 | 0.00 | 0.00 | 0.00 | 22,800,000.00 *gal* |
| Application event totals | | 58.78 | 9.11 | 72.69 | 306.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 20.43 | 0.00 | 0.00 | 0.00 | 23,280,000.00 *gal* |
| Application event totals | | 20.43 | 0.00 | 0.00 | 0.00 | |

## Field 2 - 05/25/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 51.09 | 11.11 | 81.00 | 327.12 | 1,680,000.00 *gal* |
| Irrigation Well | Ground water | 18.96 | 0.00 | 0.00 | 0.00 | 21,600,000.00 *gal* |
| Application event totals | | 70.05 | 11.11 | 81.00 | 327.12 | |

| | | | | |
|---|---|---|---|---|
| 07/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 18.85 | 0.00 | 0.00 | 0.00 | 21,480,000.00 *gal* |
| Application event totals | | 18.85 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 41.97 | 9.13 | 66.54 | 268.71 | 1,380,000.00 *gal* |
| Irrigation Well | Ground water | 18.01 | 0.00 | 0.00 | 0.00 | 20,520,000.00 *gal* |
| Application event totals | | 59.98 | 9.13 | 66.54 | 268.71 | |

| | | | | |
|---|---|---|---|---|
| 08/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 17.91 | 0.00 | 0.00 | 0.00 | 20,400,000.00 *gal* |
| Application event totals | | 17.91 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 08/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 41.97 | 9.13 | 66.54 | 268.71 | 1,380,000.00 *gal* |
| Irrigation Well | Ground water | 18.43 | 0.00 | 0.00 | 0.00 | 21,000,000.00 *gal* |
| Application event totals | | 60.40 | 9.13 | 66.54 | 268.71 | |

## Field 3 - 10/13/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/13/2022

## Field 3 - 10/13/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/26/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 109.33 | 18.41 | 288.40 | 1,069.50 | 720,000.00 *gal* |
| Irrigation Well | Ground water | 9.77 | 0.00 | 0.00 | 0.00 | 2,928,000.00 *gal* |
| Application event totals | | 119.10 | 18.41 | 288.40 | 1,069.50 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 9.61 | 0.00 | 0.00 | 0.00 | 2,880,000.00 *gal* |
| Application event totals | | 9.61 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 104.67 | 24.59 | 196.27 | 827.16 | 840,000.00 *gal* |
| Irrigation Well | Ground water | 9.53 | 0.00 | 0.00 | 0.00 | 2,856,000.00 *gal* |
| Application event totals | | 114.20 | 24.59 | 196.27 | 827.16 | |

## Field 3 - 05/25/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/25/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/08/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 105.12 | 33.30 | 92.11 | 0.00 | 188.00 *ton* |
| Application event totals | | 105.12 | 33.30 | 92.11 | 0.00 | |

## Field 3 - 05/25/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 44.86 | 10.54 | 84.12 | 354.50 | 360,000.00 *gal* |
| Irrigation Well | Ground water | 15.86 | 0.00 | 0.00 | 0.00 | 4,752,000.00 *gal* |
| Application event totals | | 60.72 | 10.54 | 84.12 | 354.50 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 13.94 | 0.00 | 0.00 | 0.00 | 4,176,000.00 *gal* |
| Application event totals | | 13.94 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 39.42 | 8.57 | 62.49 | 252.35 | 240,000.00 *gal* |
| Irrigation Well | Ground water | 14.26 | 0.00 | 0.00 | 0.00 | 4,272,000.00 *gal* |
| Application event totals | | 53.68 | 8.57 | 62.49 | 252.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 12.98 | 0.00 | 0.00 | 0.00 | 3,888,000.00 *gal* |
| Application event totals | | 12.98 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 13.14 | 0.00 | 0.00 | 0.00 | 3,936,000.00 *gal* |
| Application event totals | | 13.14 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 12.82 | 0.00 | 0.00 | 0.00 | 3,840,000.00 *gal* |
| Application event totals | | 12.82 | 0.00 | 0.00 | 0.00 | |

Field 3 - 05/25/2023: Corn, silage

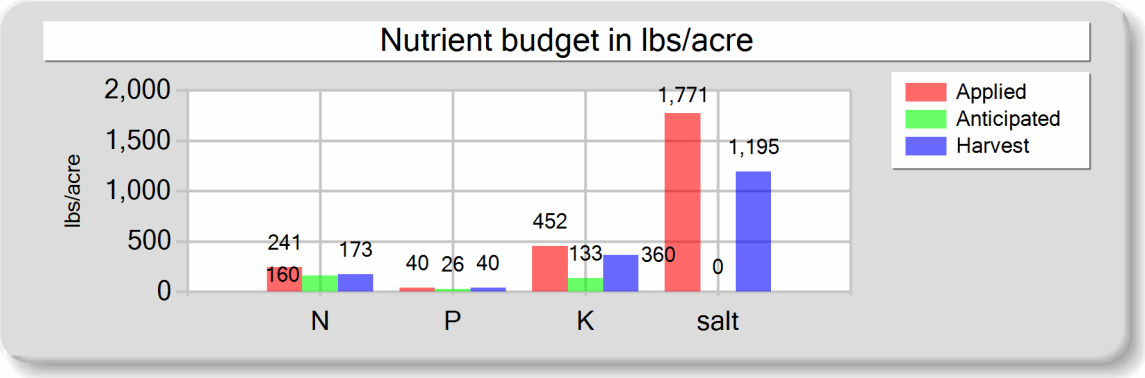
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 13.14 | 0.00 | 0.00 | 0.00 | 3,936,000.00 *gal* |
| Application event totals | | 13.14 | 0.00 | 0.00 | 0.00 | |

## B. NUTRIENT BUDGET

Field 1 - 10/13/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/13/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 200.29 | 40.40 | 451.85 | 1,770.77 |
| Fresh water | 33.60 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 240.89 | 40.40 | 451.85 | 1,770.77 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 173.20 | 39.64 | 360.46 | 1,194.97 |
| Nutrient balance | 67.69 | 0.76 | 91.39 | 575.80 |
| Applied to removed ratio | 1.39 | 1.02 | 1.25 | 1.48 |

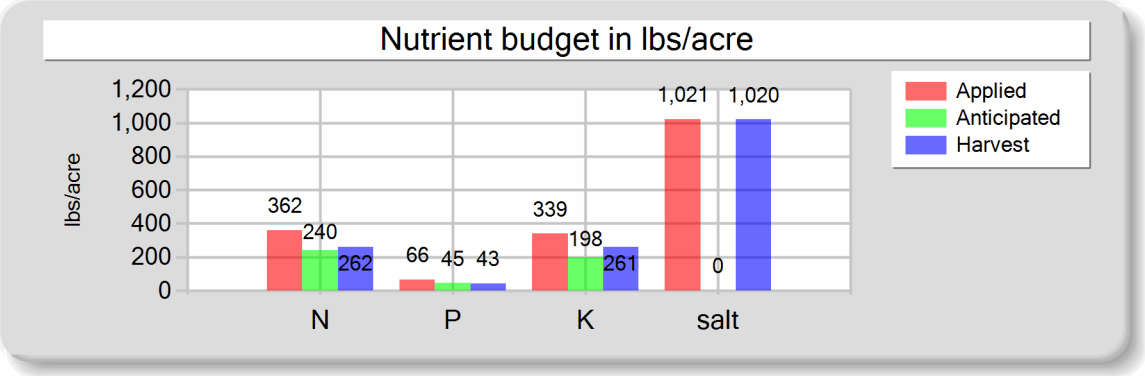
| Fresh water applied | |
|---|---|
| 56,928,000.00 | *gallons* |
| 2,096.46 | *acre-inches* |
| 12.78 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 9,600,000.00 | *gallons* |
| 353.54 | *acre-inches* |
| 2.16 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 1 - 05/25/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/25/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 104.84 | 33.22 | 91.86 | 0.00 |
| Process wastewater | 143.49 | 32.40 | 247.30 | 1,021.24 |
| Fresh water | 106.19 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 361.51 | 65.61 | 339.16 | 1,021.24 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 261.72 | 42.82 | 260.76 | 1,019.94 |
| Nutrient balance | 99.79 | 22.80 | 78.40 | 1.30 |
| Applied to removed ratio | 1.38 | 1.53 | 1.30 | 1.00 |

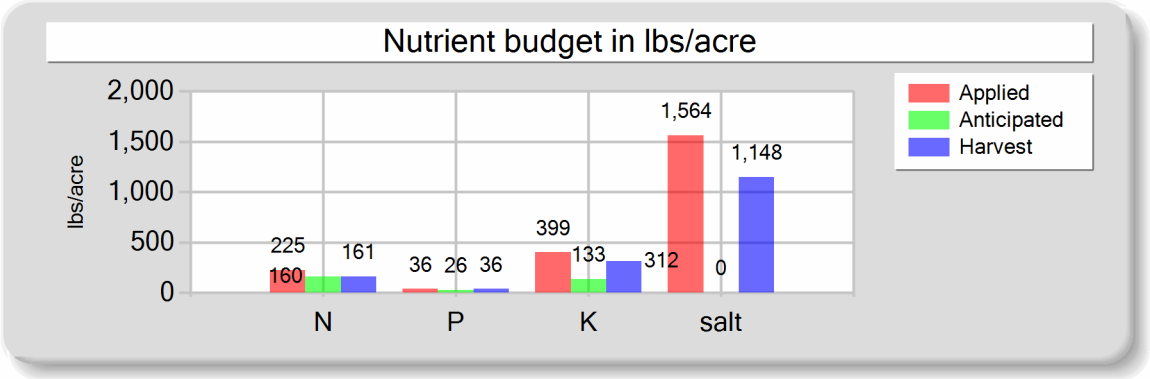
| Fresh water applied | |
|---|---|
| 179,904,000.00 | *gallons* |
| 6,625.25 | *acre-inches* |
| 40.40 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,600,000.00 | *gallons* |
| 243.06 | *acre-inches* |
| 1.48 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 10/13/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/13/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 176.65 | 35.55 | 399.45 | 1,564.04 |
| Fresh water | 41.08 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 224.73 | 35.55 | 399.45 | 1,564.04 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 160.90 | 36.49 | 312.24 | 1,148.47 |
| Nutrient balance | 63.82 | -0.94 | 87.21 | 415.57 |
| Applied to removed ratio | 1.40 | 0.97 | 1.28 | 1.36 |

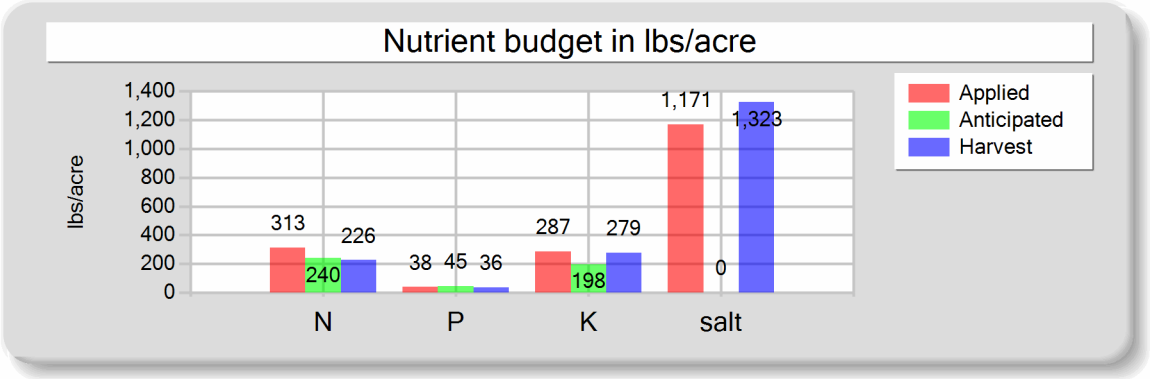
| Fresh water applied | |
|---|---|
| 46,800,000.00 | *gallons* |
| 1,723.48 | *acre-inches* |
| 12.77 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,960,000.00 | *gallons* |
| 256.31 | *acre-inches* |
| 1.90 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 05/25/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/25/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 173.80 | 38.47 | 286.77 | 1,170.90 |
| Fresh water | 132.61 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 313.41 | 38.47 | 286.77 | 1,170.90 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 226.44 | 36.34 | 278.63 | 1,323.26 |
| Nutrient balance | 86.97 | 2.13 | 8.14 | -152.36 |
| Applied to removed ratio | 1.38 | 1.06 | 1.03 | 0.88 |

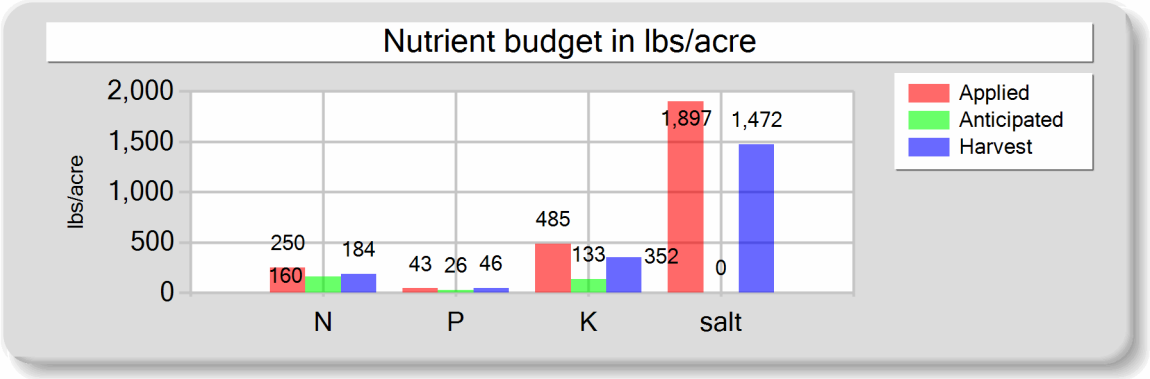
| Fresh water applied | |
|---|---|
| 151,080,000.00 | *gallons* |
| 5,563.76 | *acre-inches* |
| 41.21 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,120,000.00 | *gallons* |
| 225.38 | *acre-inches* |
| 1.67 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 10/13/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/13/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 214.00 | 43.00 | 484.68 | 1,896.65 |
| Fresh water | 28.92 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 249.92 | 43.00 | 484.68 | 1,896.65 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 183.97 | 45.62 | 351.75 | 1,471.76 |
| Nutrient balance | 65.95 | -2.62 | 132.93 | 424.89 |
| Applied to removed ratio | 1.36 | 0.94 | 1.38 | 1.29 |

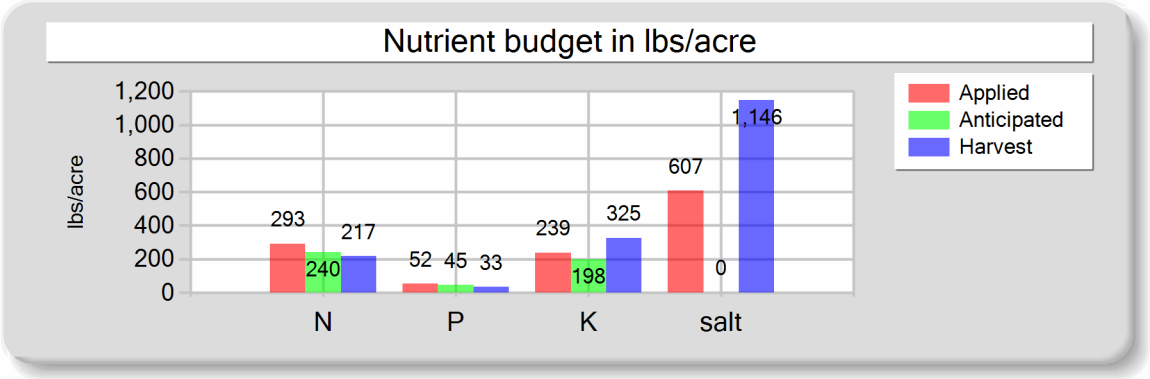
| Fresh water applied | |
|---|---|
| 8,664,000.00 | *gallons* |
| 319.07 | *acre-inches* |
| 12.76 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,560,000.00 | *gallons* |
| 57.45 | *acre-inches* |
| 2.30 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 05/25/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/25/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 105.12 | 33.30 | 92.11 | 0.00 |
| Process wastewater | 84.27 | 19.11 | 146.60 | 606.85 |
| Fresh water | 96.13 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 292.53 | 52.42 | 238.71 | 606.85 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 217.02 | 32.99 | 324.66 | 1,145.85 |
| Nutrient balance | 75.51 | 19.43 | -85.94 | -539.00 |
| Applied to removed ratio | 1.35 | 1.59 | 0.74 | 0.53 |

| Fresh water applied | |
|---|---|
| 28,800,000.00 | *gallons* |
| 1,060.61 | *acre-inches* |
| 42.42 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 600,000.00 | *gallons* |
| 22.10 | *acre-inches* |
| 0.88 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Corral Manure

Sample and source description: Corral Manure

Sample date: 05/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 20,200.00 | 6,400.00 | 17,700.00 | | | | | | | 0.00 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | 1.00 |

### Corral Manure

Sample and source description: Corral Manure

Sample date: 11/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 23,300.00 | 6,500.00 | 44,800.00 | | | | | | | 0.00 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | 1.00 |

## B. PROCESS WASTEWATER ANALYSES

### 2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 453.00 | 286.00 | 0.00 | 1.90 | 76.60 | 1,200.00 | | | | | | | | 8,680.00 | 4,450 |
| **DL** | 0.01 | 0.01 | 0.10 | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

## Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 282.00 | 166.00 | 0.00 | 0.70 | 65.90 | 543.00 | | | | | | | | 4,360.00 | 2,200 |
| **DL** | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

## Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 371.00 | 269.00 | 0.00 | 2.30 | 87.70 | 700.00 | | | | | | | | 6,340.00 | 2,950 |
| **DL** | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

## Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 491.00 | 330.00 | 0.00 | 1.00 | 107.00 | 780.00 | | | | | | | | 6,620.00 | 3,150 |
| **DL** | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

## Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 422.00 | 324.00 | 0.00 | 0.00 | 107.00 | 878.00 | | | | | | | | 7,240.00 | 3,800 |
| **DL** | 0.70 | 0.20 | 0.01 | 0.20 | 0.02 | 0.02 | | | | | | | | 2.00 | 10 |

## C. FRESH WATER ANALYSES

### Domestic Well

#### Domestic Well Milk Barn

Sample description: Domestic Well Milk Barn

Sample date: 12/06/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 5.50 | 0.00 | 5.50 | | | | | | | | 431.00 | 0 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 1.00 | 1 |

### Irrigation Well

#### Irrigation Well #1

Sample description: Irrigation Well #1

Sample date: 11/09/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 11.60 | 0.00 | 11.60 | | | | | | | | 742.00 | 0 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

#### Irrigation Well #2

Sample description: Irrigation Well #2

Sample date: 11/09/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 14.20 | 0.00 | 14.20 | | | | | | | | 903.00 | 0 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

Irrigation Well

Irrigation Well #3

Sample description: Irrigation Well #3

Sample date: 11/09/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 12.40 | 0.00 | 12.40 | | | | | | | | 803.00 | 0 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

Irrigation Well #4

Sample description: Irrigation Well #4

Sample date: 11/09/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 10.00 | 0.00 | 10.00 | | | | | | | | 682.00 | 0 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 1 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

Field 1 - 10/13/2022: Oats, silage-soft dough

Field 1 Oat Silage A

Sample and source description: Field 1 Oat Silage A

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,100.00 | 2,200.00 | 20,800.00 | | 7.60 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 1 - 10/13/2022: Oats, silage-soft dough

### Field 1 Oat Silage B

Sample and source description: Field 1 Oat Silage B

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,900.00 | 2,300.00 | 21,600.00 | | 8.10 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

### Field 1 Oat Silage C

Sample and source description: Field 1 Oat Silage C

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,300.00 | 2,200.00 | 21,700.00 | | 6.90 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

### Field 1 Oat Silage D

Sample and source description: Field 1 Oat Silage D

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,000.00 | 3,800.00 | 32,600.00 | | 10.00 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 1 - 10/13/2022: Oats, silage-soft dough

### Field 1 Oat Silage E

Sample and source description: Field 1 Oat Silage E

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,300.00 | 3,600.00 | 31,500.00 | | 9.70 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 1 - 05/25/2023: Corn, silage

### Field 1 Corn Silage A

Sample and source description: Field 1 Corn Silage A

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,600.00 | 2,000.00 | 10,900.00 | | 5.10 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

### Field 1 Corn Silage B

Sample and source description: Field 1 Corn Silage B

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 16,100.00 | 3,000.00 | 15,600.00 | | 5.90 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 1 - 05/25/2023: Corn, silage

### Field 1 Corn Silage C

Sample and source description: Field 1 Corn Silage C

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,600.00 | 1,600.00 | 14,300.00 | | 4.90 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

### Field 1 Corn Silage D

Sample and source description: Field 1 Corn Silage D

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,100.00 | 2,300.00 | 13,400.00 | | 5.20 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 2 - 10/13/2022: Oats, silage-soft dough

### Field 2 Oat Silage A

Sample and source description: Field 2 Oat Silage A

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 14,100.00 | 3,300.00 | 27,000.00 | | 10.10 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 2 - 10/13/2022: Oats, silage-soft dough

### Field 2 Oat Silage B

Sample and source description: Field 2 Oat Silage B

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,800.00 | 2,800.00 | 25,200.00 | | 9.00 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 2 - 05/25/2023: Corn, silage

### Field 2 Corn Silage A

Sample and source description: Field 2 Corn Silage A

Sample date: 09/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,300.00 | 2,300.00 | 13,900.00 | | 7.50 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

### Field 2 Corn Silage B

Sample and source description: Field 2 Corn Silage B

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,000.00 | 1,600.00 | 16,000.00 | | 6.60 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## Field 3 - 10/13/2022: Oats, silage-soft dough

Field 3 - 10/13/2022: Oats, silage-soft dough

Field 3 Oat Silage

Sample and source description: Field 3 Oat Silage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,500.00 | 3,100.00 | 23,900.00 | | 10.00 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

Field 3 - 05/25/2023: Corn, silage

Field 3 Corn Silage

Sample and source description: Field 3 Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,500.00 | 1,900.00 | 18,700.00 | | 6.60 |
| **DL** | 0.01 | 0.01 | 0.01 | | 1.00 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

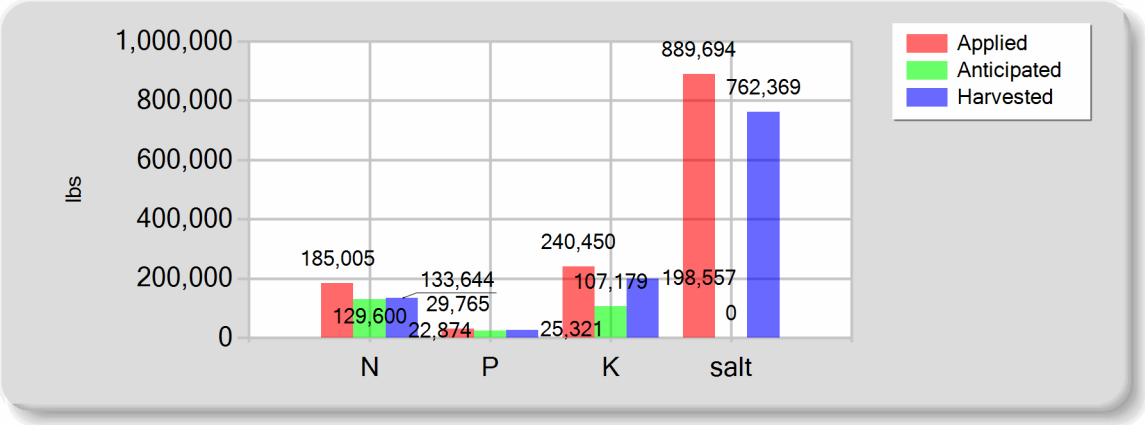
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

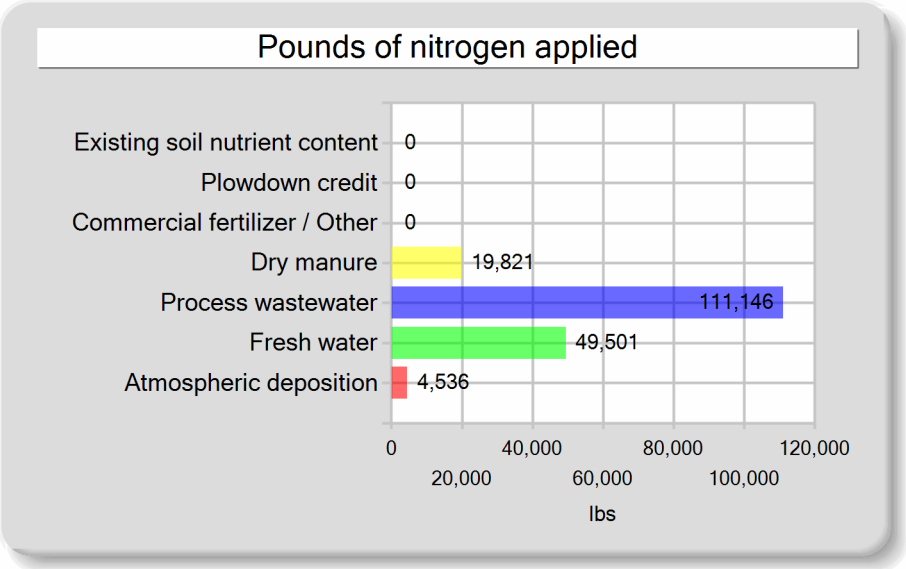
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

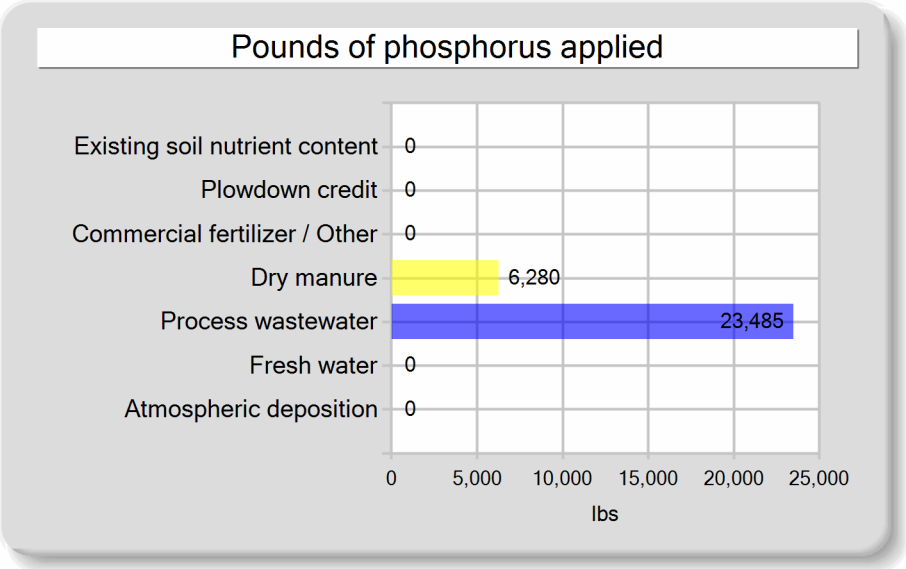
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 19,821.37 | 6,280.04 | 17,368.23 | 0.00 |
| Process wastewater | 111,146.39 | 23,485.08 | 223,081.88 | 889,693.83 |
| Fresh water | 49,500.76 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 4,536.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 185,004.52 | 29,765.12 | 240,450.11 | 889,693.83 |
| Anticipated crop nutrient removal | 129,600.00 | 22,874.40 | 107,179.20 | 0.00 |
| Actual crop nutrient removal | 133,644.00 | 25,321.40 | 198,557.25 | 762,368.55 |
| Nutrient balance | 51,360.52 | 4,443.73 | 41,892.86 | 127,325.28 |
| Applied to removed ratio | 1.38 | 1.18 | 1.21 | 1.17 |

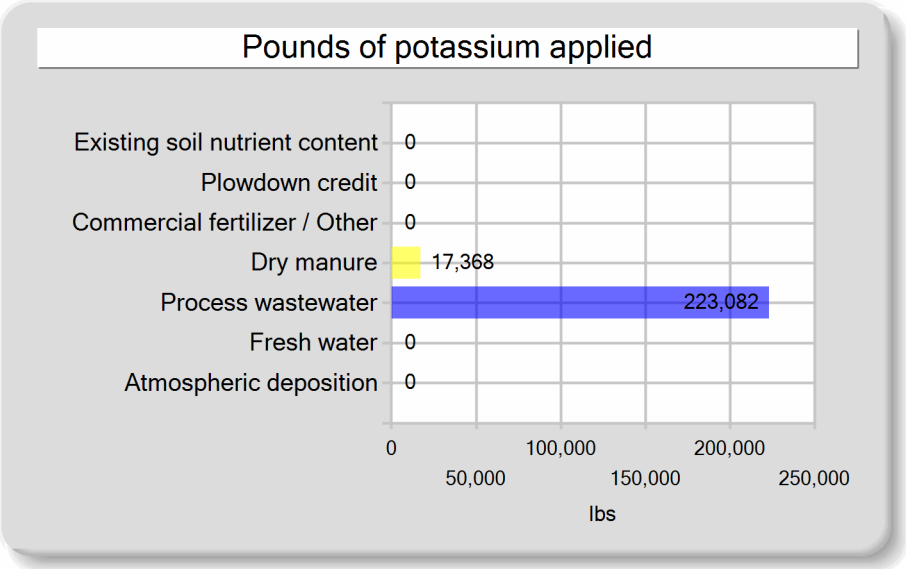
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

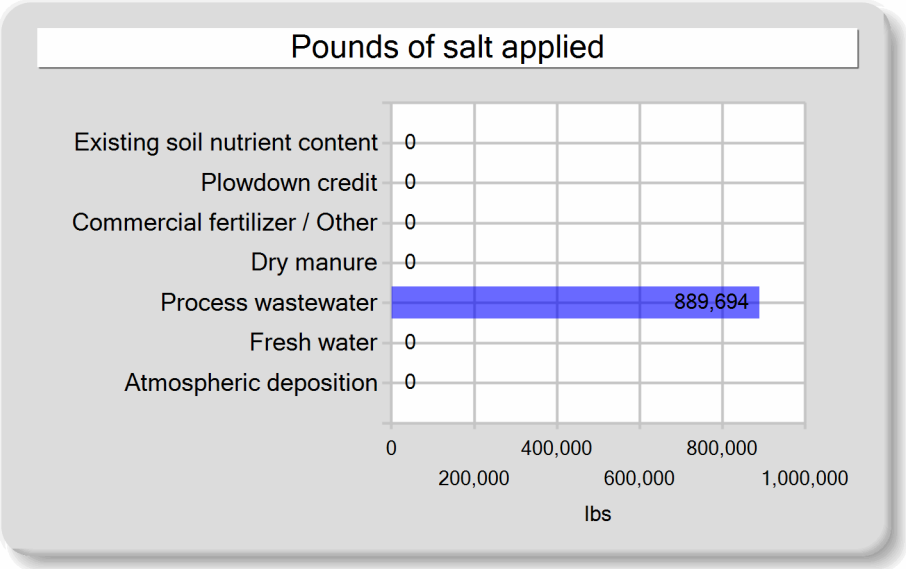


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

*No notes entered for this annual report.*

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Joe Louie Bettencourt | Joe Alves |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| | |
| DATE | DATE |

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

DELLAVALLE LABORATORY INC

# Report of Dairy Well Water Analysis

Alves & Son Dairy #2
13745 Newport Rd
Turlock CA 95380
00-0023676 08

**E-mail:**
**Copy To:** cas.labs@yahoo.com

**Lab No.:** 23K0518
**Sampled By:** M Dekeyrel
**Requested By:** Melissa Alves
**Submitted Date:** 11/09/23
**Reported Date:** 11/16/23
**Project:**
**Crop ID:**

| | | Date Sampled | Time Sampled | EC µmhos/cm | EC mmhos/cm | $NO_3$-N mg/L | Field $NH_4$-N mg/L | Total $NH_4$-N mg/L | pH at 25°C unit |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Irrigation Well #1 | 11/09/23 | 12:20 | 742 | 0.74 | 11.6 | ND | * | 7.8 |
| 2 | Irrigation Well #2 | 11/09/23 | 12:25 | 903 | 0.90 | 14.2 | ND | * | 7.6 |
| 3 | Irrigation Well #3 | 11/09/23 | 12:40 | 803 | 0.80 | 12.4 | ND | * | 7.7 |
| 4 | Irrigation Well #4 | 11/09/23 | 12:35 | 682 | 0.68 | 10.0 | ND | * | 7.8 |

***** = Field $NH_4$-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

# Report of Dairy Well Water Analysis

Alves & Son Dairy #2
13745 Newport Rd
Turlock CA 95380
00-0023676 08

**E-mail:**
**Copy To:** cas.labs@yahoo.com

**Lab No.:** 23K0518
**Sampled By:** M Dekeyrel
**Requested By:** Melissa Alves
**Submitted Date:** 11/09/23
**Reported Date:** 11/16/23
**Project:**
**Crop ID:**

| | | Date Sampled | Time Sampled | EC µmhos/cm | EC mmhos/cm | $NO_3$-N mg/L | Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot | lbs/AcInch |
|---|---|---|---|---|---|---|---|---|
| 1 | Irrigation Well #1 | 11/09/23 | 12:20 | 742 | 0.74 | **11.6** | 31.67 | 2.64 |
| 2 | Irrigation Well #2 | 11/09/23 | 12:25 | **903** | **0.90** | **14.2** | 38.77 | 3.23 |
| 3 | Irrigation Well #3 | 11/09/23 | 12:40 | 803 | 0.80 | **12.4** | 33.85 | 2.82 |
| 4 | Irrigation Well #4 | 11/09/23 | 12:35 | 682 | 0.68 | **10.0** | 27.30 | 2.28 |

| **Water for Crop Use** **Nutrient Management** | Total Salts | Total Salts | Nitrate Nitrogen |
|---|---|---|---|
| **Low Levels** | < 900 | < 0.90 | < 2.0 |
| **Significant Levels** | 900-2200 | 0.90-2.2 | 2.0-10.0 |
| **High Levels** | 2200+ | 2.2+ | 10.0+ |

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
**MDL** = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23K0518-01 | Irrigation Well #1 | Ag Water | M Dekeyrel | | 11/09/2023 12:20 |
| 23K0518-02 | Irrigation Well #2 | Ag Water | M Dekeyrel | | 11/09/2023 12:25 |
| 23K0518-03 | Irrigation Well #3 | Ag Water | M Dekeyrel | | 11/09/2023 12:40 |
| 23K0518-04 | Irrigation Well #4 | Ag Water | M Dekeyrel | | 11/09/2023 12:35 |

Default Cooler Temperature on Receipt °C: 0.1
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

*The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.*

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Sample Results

**Sample: Irrigation Well #1**
**23K0518-01 (Water)**

Sampled: 11/9/2023 12:20
Sampled By: M Dekeyrel

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.74** | mmhos/cm | 0.01 | 1 | | 11/15/23 11:37 | SM 2510 B | | BEK0350 |
| **Electrical Conductivity umhos** | **742** | umhos/cm | 10.0 | 1 | | 11/15/23 11:37 | SM 2510 B | | BEK0350 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/09/23 12:20 | Field | | BEK0316 |
| **Nitrate Nitrogen as NO3N** | **11.6** | mg/L | 0.1 | 1 | 10 | 11/10/23 16:33 | EPA 300.0 | | BEK0340 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/15/23 11:37 | SM 4500-H+ | H | BEK0350 |
| **pH** | **7.8** | units | 1.0 | 1 | | 11/15/23 11:37 | SM 4500-H+ | H | BEK0350 |

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LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Sample Results
**(Continued)**

**Sample: Irrigation Well #2**
**23K0518-02 (Water)**

Sampled: 11/9/2023 12:25
Sampled By: M Dekeyrel

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.90** | mmhos/cm | 0.01 | 1 | | 11/15/23 11:39 | SM 2510 B | | BEK0350 |
| **Electrical Conductivity umhos** | **903** | umhos/cm | 10.0 | 1 | | 11/15/23 11:39 | SM 2510 B | | BEK0350 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/09/23 12:25 | Field | | BEK0316 |
| **Nitrate Nitrogen as NO3N** | **14.2** | mg/L | 0.1 | 1 | 10 | 11/10/23 16:54 | EPA 300.0 | | BEK0340 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/15/23 11:39 | SM 4500-H+ | H | BEK0350 |
| **pH** | **7.6** | units | 1.0 | 1 | | 11/15/23 11:39 | SM 4500-H+ | H | BEK0350 |

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DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Sample Results
**(Continued)**

**Sample: Irrigation Well #3**
**23K0518-03 (Water)**

Sampled: 11/9/2023 12:40
Sampled By: M Dekeyrel

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.80** | mmhos/cm | 0.01 | 1 | | 11/15/23 11:40 | SM 2510 B | | BEK0350 |
| **Electrical Conductivity umhos** | **803** | umhos/cm | 10.0 | 1 | | 11/15/23 11:40 | SM 2510 B | | BEK0350 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/09/23 12:40 | Field | | BEK0316 |
| **Nitrate Nitrogen as NO3N** | **12.4** | mg/L | 0.1 | 1 | 10 | 11/10/23 17:14 | EPA 300.0 | | BEK0340 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/15/23 11:40 | SM 4500-H+ | H | BEK0350 |
| **pH** | **7.7** | units | 1.0 | 1 | | 11/15/23 11:40 | SM 4500-H+ | H | BEK0350 |

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LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Sample Results
**(Continued)**

**Sample: Irrigation Well #4**
**23K0518-04 (Water)**

Sampled: 11/9/2023 12:35
Sampled By: M Dekeyrel

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.68** | mmhos/cm | 0.01 | 1 | | 11/15/23 11:42 | SM 2510 B | | BEK0350 |
| **Electrical Conductivity umhos** | **682** | umhos/cm | 10.0 | 1 | | 11/15/23 11:42 | SM 2510 B | | BEK0350 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/09/23 12:35 | Field | | BEK0316 |
| **Nitrate Nitrogen as NO3N** | **10.0** | mg/L | 0.1 | 1 | 10 | 11/10/23 21:18 | EPA 300.0 | | BEK0340 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/15/23 11:42 | SM 4500-H+ | H | BEK0350 |
| **pH** | **7.8** | units | 1.0 | 1 | | 11/15/23 11:42 | SM 4500-H+ | H | BEK0350 |

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DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0340*** | | | | | | | | | | |
| **Blank (BEK0340-BLK1)** | | | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0340-BLK2)** | | | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0340-BLK3)** | | | | | Prepared & Analyzed: 11/11/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEK0340-BS1)** | | | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.1 | | 0.1 | mg/L | 5.000 | | 103 | 90-110 | | |
| **LCS (BEK0340-BS2)** | | | | | Prepared & Analyzed: 11/11/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.2 | | 0.1 | mg/L | 5.000 | | 103 | 90-110 | | |
| **Duplicate (BEK0340-DUP1)** | | **Source: 23K0518-01** | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 11.6 | | 0.1 | mg/L | | 11.6 | | | 0.423 | 10 |
| **Duplicate (BEK0340-DUP2)** | | **Source: 23K0518-04** | | | Prepared & Analyzed: 11/11/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.4 | | 0.1 | mg/L | | 10.0 | | | 3.40 | 10 |
| **Matrix Spike (BEK0340-MS1)** | | **Source: 23K0518-01** | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 16.9 | | 0.1 | mg/L | 5.000 | 11.6 | 106 | 90-110 | | |
| **Matrix Spike (BEK0340-MS2)** | | **Source: 23K0518-04** | | | Prepared & Analyzed: 11/11/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 15.9 | | 0.1 | mg/L | 5.000 | 10.0 | 117 | 90-110 | | |
| **Reference (BEK0340-SRM1)** | | | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.2 | | | mg/L | 10.00 | | 102 | 90-110 | | |
| **Reference (BEK0340-SRM2)** | | | | | Prepared & Analyzed: 11/10/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BEK0340-SRM3)** | | | | | Prepared & Analyzed: 11/11/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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DELLAVALLE™ LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0350*** | | | | | | | | | | |
| **Blank (BEK0350-BLK1)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.6 | | 1.0 | units | | | | | | |
| **Blank (BEK0350-BLK2)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.6 | | 1.0 | units | | | | | | |
| **Blank (BEK0350-BLK3)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.6 | | 1.0 | units | | | | | | |
| **Duplicate (BEK0350-DUP1)** | | **Source: 23K0483-01** | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 3.23 | | 0.01 | mmhos/cm | | 3.27 | | | 1.28 | 10 |
| pH | 7.4 | | 1.0 | units | | 7.4 | | | 0.269 | 10 |
| Electrical Conductivity umhos | 3230 | | 10.0 | umhos/cm | | 3270 | | | 1.28 | 10 |
| **Duplicate (BEK0350-DUP2)** | | **Source: 23K0518-04** | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 0.68 | | 0.01 | mmhos/cm | | 0.68 | | | 0.337 | 10 |
| Electrical Conductivity umhos | 685 | | 10.0 | umhos/cm | | 682 | | | 0.337 | 10 |
| pH | 7.8 | | 1.0 | units | | 7.8 | | | 0.256 | 10 |
| **Reference (BEK0350-SRM1)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 576 | | | umhos/cm | 538.0 | | 107 | 90-110 | | |
| **Reference (BEK0350-SRM2)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| pH | 7.5 | | | units | 7.520 | | 100 | 67021-101.32 | | |
| **Reference (BEK0350-SRM3)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| Electrical Conductivity umhos | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| **Reference (BEK0350-SRM4)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| Electrical Conductivity umhos | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| **Reference (BEK0350-SRM5)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |

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DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:37

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0350 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0350-SRM5)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| Electrical Conductivity umhos | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |
| **Reference (BEK0350-SRM6)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEK0350-SRM7)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEK0350-SRM8)** | | | | | Prepared: 11/10/2023 | Analyzed: 11/15/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |

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11/09/23 16:20 23K0518

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

## WATER WORK REQUEST

Acct No. 11-9-23 Tests.

Bill To: 23676 8

Purchase Order No. Results Needed By

No. of Samples 4 No. Bottles 4

**Water Type:** [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Client **Alves & Son Dairy #2**
Address 13745 Newport Road
City, State, Zip Turlock CA 95380
Phone Fax
Cell/Email

Copy to Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by Melissa Alves - (209) 585-8221

Ranch

Date sampled 11/9/23

Sampled by mDeKuyel

[ X ] QA/QC Document [ X ] Copy of Chain [ ] RWQCB

**Analysis and Bottles Required:** *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, $NO_3$-N, **$NH_4$-N Field Test**)
*(1) l L plastic, unpreserved (white)*

( ) DWW2: (DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS)
*(1) l L plastic, unpreserved (white)*

( ) DCW1: (EC, $NO_3$-N, TDS)
*(1) l L plastic, unpreserved (white)*

( ) DPW1: (EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK )
*(1) l L plastic, unpreserved (white)*

( ) DPW2: (DPW1 Plus Ca, Mg, Na, $HCO_3$, $CO_3$, $SO_4$, Cl)
*(1) l L plastic, unpreserved (white)*

( ) Other

| DESCRIPTION OF SAMPLES | | Date Sampled | Time Sampled | Field NH4-N (mg/L) | Received Temp °C |
|---|---|---|---|---|---|
| 1. Irrigation well #1 | Sampled From: | 11/9/23 | 1220p | Ø | -0.1 |
| 2. Irrigation well #2 | Sampled From: | 11/9/23 | 1225p | Ø | -1.1 |
| 3. Irrigation well #3 | Sampled From: | 11/9/23 | 1240p | Ø | -1.1 |
| 4. Irrigation well #4 | Sampled From: | 11/9/23 | 1235p | Ø | -0.5 |
| 5. | Sampled From: | | | | |
| 6. | Sampled From: | | | | |
| 7. | Sampled From: | | | | |
| 8. | Sampled From: | | | | |
| 9. | Sampled From: | | | | |
| 10. | Sampled From: | | | | |

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | mDeKuyel | CASINC | 11/9/23 1240p | 11/9/23 1pm |
| Second | JCF | DLS | 11-9-23 1:00 | 11-9-23 3:55 |
| Third | | | | |
| Fourth | [illegible] | DCI | 11/9 16:20 | |

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

*Invoicing Information:*

*Shipping*
$ In
$ Out

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid Rec By Check No. Date

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

**Shipping Information:** Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐

**Container:** Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

**Samples Preserved with $HNO_3$ or $H_2SO_4$ were:** ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

11/09/23 16:20 23K0518

| Type of Container(s) Received | | Sample Number 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sample Containers for Internal (DLI) Use** *(Containers that go into the Lab)* | | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| | * pH Value | | | | | | | | | | |
| | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| | * pH Value | | | | | | | | | | |
| | 500 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 1 L unpreserved (White) Plastic | 1 | 1 | 1 | 1 | | | | | | |
| | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| " | 500mL unpreserved (White) Glass | | | | | | | | | | |
| | PO4-P Kit | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |
| **Sample Containers for Subcontracted ("Send Out") Analyses** *(Containers that go in the Subcontract ("Send Out") Refrigerator)* | | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| | 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| | 1 L unpreserved (White) Plastic | | | | | | | | | | |
| | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| | 1 L $HNO_3$ (Red) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| | 40 mL AG VOA, $Na_2S_2O_3$ | | | | | | | | | | |
| | 40mL AG VOA unpreserved (White) (Set of 3 | | | | | | | | | | |
| | 40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3) | | | | | | | | | | |
| | 40mL VOA, $H_3PO_4$ (Set of 3) | | | | | | | | | | |
| | 40 mL VOA, HCl (Blue) (Set of 3) | | | | | | | | | | |
| | 40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3) | | | | | | | | | | |
| Glass | 250 mL AG unpreserved (White) | | | | | | | | | | |
| | 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| | 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| | 500 mL glass unpreserved (White) | | | | | | | | | | |
| | 500 mL AG HCl (Blue) | | | | | | | | | | |
| | 1 L AG unpreserved (White) | | | | | | | | | | |
| | 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| | 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 1 L AG HCl (Blue) | | | | | | | | | | |
| Special | $Cr^{6+}$ - 50mL Plastic w/Borate/$HCO_3$/$CO_3$ | | | | | | | | | | |
| | Cyanide - 500 mL NaOH | | | | | | | | | | |
| | Asbestos - 1L P wrapped in foil (Set of 2) | | | | | | | | | | |
| | Sulfide - 1 L **AG** or **P** NaOH + ZnAc | | | | | | | | | | |
| | Chlorite/Bromate - 250 mL AG with EDA | | | | | | | | | | |
| | HAA5 - 250mL AG Ammonium Chlorite | | | | | | | | | | |
| | DO KIT | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |

SEND OUT

# Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock CA 95380
00-0023676 08

**E-mail:**
**Copy To:** cas.labs@yahoo.com

**Lab No.:** 23L0455
**Sampled By:** M Dekeyrel
**Requested By:** Melissa Alves
**Submitted Date:** 12/07/23
**Reported Date:** 12/13/23
**Project:**
**Crop ID:**

| | | Date Sampled | Time Sampled | EC µmhos/cm | EC mmhos/cm | $NO_3$-N mg/L | Field $NH_4$-N mg/L | Total $NH_4$-N mg/L | pH at 25°C unit |
|---|---|---|---|---|---|---|---|---|---|
| 1 | DW - MB | 12/06/23 | 11:30 | 431 | 0.43 | 5.5 | ND | | 7.7 |

***** = Field $NH_4$-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

# Report of Dairy Well Water Analysis

Alves & Son Dairy #2
13745 Newport Rd
Turlock CA 95380
00-0023676 08

**E-mail:**
**Copy To:** cas.labs@yahoo.com

**Lab No.:** 23L0455
**Sampled By:** M Dekeyrel
**Requested By:** Melissa Alves
**Submitted Date:** 12/07/23
**Reported Date:** 12/13/23
**Project:**
**Crop ID:**

| | | Date Sampled | Time Sampled | EC µmhos/cm | EC mmhos/cm | $NO_3$-N mg/L | Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot | lbs/AcInch |
|---|---|---|---|---|---|---|---|---|
| 1 | DW - MB | 12/06/23 | 11:30 | 431 | 0.43 | **5.5** | 15.02 | 1.25 |

| **Water for Crop Use** **Nutrient Management** | Total Salts | Total Salts | Nitrate Nitrogen |
|---|---|---|---|
| **Low Levels** | < 900 | < 0.90 | < 2.0 |
| **Significant Levels** | 900-2200 | 0.90-2.2 | 2.0-10.0 |
| **High Levels** | 2200+ | 2.2+ | 10.0+ |

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
**MDL** = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #2
13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:27

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23L0455-01 | DW - MB | Ag Water | M Dekeyrel | | 12/06/2023 11:30 |

Default Cooler Temperature on Receipt °C: -2.9
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Received: 12/07/2023 15:35
Reported: 12/13/2023 15:27

## Sample Results

**Sample: DW - MB**
**23L0455-01 (Water)**

Sampled: 12/6/2023 11:30
Sampled By: M Dekeyrel

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.43** | mmhos/cm | 0.01 | 1 | | 12/11/23 17:22 | SM 2510 B | | BEL0302 |
| **Electrical Conductivity umhos** | **431** | umhos/cm | 10.0 | 1 | | 12/11/23 17:22 | SM 2510 B | | BEL0302 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 12/06/23 11:30 | Field | | BEL0484 |
| **Nitrate Nitrogen as NO3N** | **5.5** | mg/L | 0.1 | 1 | 10 | 12/08/23 07:34 | EPA 300.0 | | BEL0227 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 12/11/23 17:22 | SM 4500-H+ | H | BEL0302 |
| **pH** | **7.7** | units | 1.0 | 1 | | 12/11/23 17:22 | SM 4500-H+ | H | BEL0302 |

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Reported: 12/13/2023 15:27

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEL0227*** | | | | | | | | | | |
| **Blank (BEL0227-BLK1)** | | | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEL0227-BLK2)** | | | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEL0227-BLK3)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEL0227-BLK4)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEL0227-BS1)** | | | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.3 | | 0.1 | mg/L | 5.000 | | 106 | 90-110 | | |
| **LCS (BEL0227-BS2)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.1 | | 0.1 | mg/L | 5.000 | | 102 | 90-110 | | |
| **LCS (BEL0227-BS3)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.1 | | 0.1 | mg/L | 5.000 | | 101 | 90-110 | | |
| **Duplicate (BEL0227-DUP1)** | | **Source: 23L0251-01** | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 2.6 | | 0.1 | mg/L | | 2.6 | | | 0.610 | 10 |
| **Duplicate (BEL0227-DUP2)** | | **Source: 23L0351-02** | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 0.04 | | 0.1 | mg/L | | 0.04 | | | 0.00 | 10 |
| **Duplicate (BEL0227-DUP3)** | | **Source: 23L0446-01** | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 3.9 | | 0.1 | mg/L | | 3.9 | | | 0.00 | 10 |
| **Matrix Spike (BEL0227-MS1)** | | **Source: 23L0251-01** | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 7.8 | | 0.1 | mg/L | 5.000 | 2.6 | 103 | 90-110 | | |
| **Matrix Spike (BEL0227-MS2)** | | **Source: 23L0351-02** | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.3 | | 0.1 | mg/L | 5.000 | 0.04 | 105 | 90-110 | | |
| **Matrix Spike (BEL0227-MS3)** | | **Source: 23L0446-01** | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.1 | | 0.1 | mg/L | 5.000 | 3.9 | 104 | 90-110 | | |
| **Reference (BEL0227-SRM1)** | | | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.1 | | | mg/L | 10.00 | | 101 | 90-110 | | |
| **Reference (BEL0227-SRM2)** | | | | | Prepared & Analyzed: 12/7/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.2 | | | mg/L | 10.00 | | 102 | 90-110 | | |

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13745 Newport Rd
Turlock, CA 95380

Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:27

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEL0227 (Continued)*** | | | | | | | | | | |
| **Reference (BEL0227-SRM3)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BEL0227-SRM4)** | | | | | Prepared & Analyzed: 12/8/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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Reported: 12/13/2023 15:27

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEL0302*** | | | | | | | | | | |
| **Blank (BEL0302-BLK1)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 5.7 | | 1.0 | units | | | | | | |
| **Blank (BEL0302-BLK2)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.4 | | 1.0 | units | | | | | | |
| **Blank (BEL0302-BLK3)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.4 | | 1.0 | units | | | | | | |
| **Duplicate (BEL0302-DUP1)** | | **Source: 23L0456-08** | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 0.74 | | 0.01 | mmhos/cm | | 0.73 | | | 1.20 | 10 |
| pH | 7.5 | | 1.0 | units | | 7.6 | | | 0.133 | 10 |
| Electrical Conductivity umhos | 739 | | 10.0 | umhos/cm | | 730 | | | 1.20 | 10 |
| **Duplicate (BEL0302-DUP2)** | | **Source: 23L0553-01** | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 5.41 | | 0.01 | mmhos/cm | | 5.37 | | | 0.668 | 10 |
| Electrical Conductivity umhos | 5410 | | 10.0 | umhos/cm | | 5370 | | | 0.668 | 10 |
| pH | 7.1 | | 1.0 | units | | 7.1 | | | 0.141 | 10 |
| **Reference (BEL0302-SRM1)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 434 | | | umhos/cm | 426.0 | | 102 | 90-110 | | |
| **Reference (BEL0302-SRM2)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| pH | 7.5 | | | units | 7.520 | | 99.9 | 67021-101.32 | | |
| **Reference (BEL0302-SRM3)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEL0302-SRM4)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEL0302-SRM5)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |

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Account# 00-0023676
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:27

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEL0302 (Continued)*** | | | | | | | | | | |
| **Reference (BEL0302-SRM5)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEL0302-SRM6)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEL0302-SRM7)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEL0302-SRM8)** | | | | | Prepared & Analyzed: 12/11/2023 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 99.8 | 97.5-102.5 | | |

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12/07/23 15:35 23L0455

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

## WATER WORK REQUEST

Bill To: Acct No. 236076 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Alves & Son Dairy #2
Address: 13745 Newport Road
City, State, Zip: Turlock CA 95380
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Melissa Alves - (209) 585-8221

Ranch _____

Date sampled: 12/6/23

Sampled by: mDekeuvel

[ X ] QA/QC Document [ X ] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

**Water Type:** [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

**Analysis and Bottles Required:** *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, $NO_3$-N, **$NH_4$-N Field Test**)
*(1) 1 L plastic, unpreserved (white)*

( ) DWW2: (DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS)
*(1) 1 L plastic, unpreserved (white)*

( ) DCW1: (EC, $NO_3$-N, TDS)
*(1) 1 L plastic, unpreserved (white)*

( ) DPW1: (EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK )
*(1) 1 L plastic, unpreserved (white)*

( ) DPW2: (DPW1 Plus Ca, Mg, Na, $HCO_3$, $CO_3$, $SO_4$, Cl)
*(1) 1 L plastic, unpreserved (white)*

( ) Other _____

| DESCRIPTION OF SAMPLES | | Date Sampled | Time Sampled | Field NH4-N (mg/L) | Received Temp °C |
|---|---|---|---|---|---|
| 1. DW-MB | Sampled From: | 12/6/23 | 1130A | Ø | -2.9 |
| 2. | Sampled From: | | | | |
| 3. | Sampled From: | | | | |
| 4. | Sampled From: | | | | |
| 5. | Sampled From: | | | | |
| 6. | Sampled From: | | | | |
| 7. | Sampled From: | | | | |
| 8. | Sampled From: | | | | |
| 9. | Sampled From: | | | | |
| 10. | Sampled From: | | | | |

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | mDekeuvel | CASINC | 12/6/23 1130A | 12/7/23 11A |
| Second | JCF | DLV | 12-7-23 11:00 | 12-7-23 3:15 |
| Third | | | | |
| Fourth | | DLI | 12-7-23 15:35 | |

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

*Invoicing Information:*

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

*Shipping*
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

**Shipping Information:** Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

**Container:** Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☐ Blue Ice ☑ None ☐

**Samples Preserved with $HNO_3$ or $H_2SO_4$ were:** ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

| Type of Container(s) Received | | Sample Number 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sample Containers for Internal (DLI) Use** *(Containers that go into the Lab)* | | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| | * pH Value | | | | | | | | | | |
| | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| | * pH Value | | | | | | | | | | |
| | 500 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 1 L unpreserved (White) Plastic | 1 | | | | | | | | | |
| | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| " | 500mL unpreserved (White) Glass | | | | | | | | | | |
| | PO4-P Kit | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |
| **Sample Containers for Subcontracted ("Send Out") Analyses** *(Containers that go in the Subcontract ("Send Out") Refrigerator)* | | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| | 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| | 1 L unpreserved (White) Plastic | | | | | | | | | | |
| | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| | 1 L $HNO_3$ (Red) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| | 40 mL AG VOA, $Na_2S_2O_3$ | | | | | | | | | | |
| | 40mL AG VOA unpreserved (White) (Set of 3 | | | | | | | | | | |
| | 40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3) | | | | | | | | | | |
| | 40mL VOA, $H_3PO_4$ (Set of 3) | | | | | | | | | | |
| | 40 mL VOA, HCl (Blue) (Set of 3) | | | | | | | | | | |
| | 40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3) | | | | | | | | | | |
| Glass | 250 mL AG unpreserved (White) | | | | | | | | | | |
| | 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| | 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| | 500 mL glass unpreserved (White) | | | | | | | | | | |
| | 500 mL AG HCl (Blue) | | | | | | | | | | |
| | 1 L AG unpreserved (White) | | | | | | | | | | |
| | 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| | 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| | 1 L AG HCl (Blue) | | | | | | | | | | |
| Special | $Cr^{6+}$ - 50mL Plastic w/Borate/$HCO_3$/$CO_3$ | | | | | | | | | | |
| | Cyanide - 500 mL NaOH | | | | | | | | | | |
| | Asbestos - 1L P wrapped in foil (Set of 2) | | | | | | | | | | |
| | Sulfide - 1 L **AG** or **P** NaOH + ZnAc | | | | | | | | | | |
| | Chlorite/Bromate - 250 mL AG with EDA | | | | | | | | | | |
| | HAA5 - 250mL AG Ammonium Chlorite | | | | | | | | | | |
| | DO KIT | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |
| | **Other:** | | | | | | | | | | |

12/07/23 15:35 23L0455

SEND OUT